Keshav Sharma

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EDUCATION

Vellore Institute of Technology — **Chennai, Tamil Nadu**

Bachelor of Technology in Computer Science (Data Science Specialisation) — *Graduation: May 2027*

SKILL SUMMARY

Languages: Python, Java, SQL, C/C++
Core Computer Science: Data Structures & Algorithms, OOP, Design Patterns, Multithreading, Problem Solving
Machine Learning & Deep Learning: Scikit-learn, PyTorch, TensorFlow, Pandas, NumPy, Feature Engineering, Model Evaluation, Anomaly Detection, Statistics & Probability
Generative AI & Agentic AI: LLMs, Generative AI, LangChain, RAG, Prompt Engineering, Google Gemini API, LLM Application Development, Tool Calling, Model Evaluation
Backend & APIs: FastAPI, Flask, REST APIs, API Integration, JSON, TCP/IP
Databases: SQL, MySQL, SQLite, MongoDB, Redis, Database Design, Query Optimization
Software Engineering & Testing: Git, GitHub, PyTest, Unit Testing, Integration Testing, Test-Driven Development (TDD)

EXPERIENCE

AI for Sustainability Virtual Intern* |LINK** — ***July – Sep, 2026

- Completed an AI-focused virtual internship covering Generative AI, Prompt Engineering, RAG, Agentic AI, and Responsible AI, with hands-on exposure to enterprise AI workflows and LLM applications.
- Applied RAG and semantic search concepts using document ingestion, embeddings, vector databases, and retrieval pipelines to improve the grounding and reliability of LLM responses.
- Built an Energy Anomaly Detector using Python, Pandas, NumPy, and Isolation Forest, detecting 9 anomalies across 6,918 test hours (0.13%), with automated email alerts, privacy safeguards, transparent scoring, and rate-limited notifications.

PROJECTS

***Docwise* |LINK**

- Built an AI-powered PDF question-answering system using Google Gemini and RAG, enabling contextual question answering over documents exceeding 200 pages.
- Developed a document processing and retrieval pipeline combining PDF extraction, chunking, embeddings, and semantic retrieval, achieving 95%+ query accuracy.
- Integrated FastAPI and LLM inference to deliver an end-to-end AI application for document understanding and conversational querying.

***Real-Time Stock Market Anomaly Detector* |LINK**

- Engineered 15+ time-series features from 500K+ NSE market observations and trained Isolation Forest and LSTM Autoencoder models to detect anomalous price and volume movements.
- Applied FinBERT sentiment analysis to 25K+ financial news records and integrated sentiment scores with market features, combining structured market data and unstructured financial text into a unified analytical pipeline.
- Benchmarked 2 anomaly-detection models using precision, recall, F1-score, and inference time, with the best model targeting ~86% precision, ~81% recall, and ~83% F1-score on the evaluation dataset.

***RedisLite* |LINK**

- Engineered a Redis-inspired in-memory key-value database in Python featuring a custom RESP protocol parser, multithreaded TCP server, thread-safe storage engine, and an O(1) LRU cache with lazy/active TTL expiration.
- Designed a four-layer architecture (TCP transport, storage engine, cache/TTL, persistence) implementing Write-Ahead Logging (WAL), atomic log compaction, and automatic crash recovery.
- Validated the system with 46 automated PyTest unit tests (TDD), benchmarking 619K SET ops/sec in-memory and 2.8K ops/sec with durable WAL persistence.

***CV-FL: Privacy-Preserving Federated Learning* |LINK**

- Developed a federated learning framework for IoT botnet detection using the N-BaIoT dataset, enabling collaborative model training across distributed clients without sharing raw data.
- Implemented hybrid model compression and lightweight verifiable aggregation using Merkle commitments and SignScore, achieving ~19.7× compression while preserving model integrity.
- Evaluated robustness against malicious clients and model tampering, achieving 100% tamper detection at ≥5% corruption and maintaining robustness with up to 30% malicious clients.

Certifications

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional |**LINK** 2026
- AWS APAC Solutions Architecture virtual experience program | **LINK** 2025